Audit Trail — MUTCD TA-10: Flagger one-lane two-way

Field	Value
MUTCD typical application	TA-10
CDOT standard sheet	S-630-1
Case	MUTCD TA-10: Flagger one-lane two-way
Taper length	100 ft (one-lane two-way taper)
Buffer space	305 ft
Device spacing — taper	20 ft
Device spacing — tangent	80 ft
Crew steps	19

Taper Length

Flagger alternating-flow: one-lane two-way taper (50-100 ft fixed band per MUTCD Sec 6B.08) -> using 100 ft maximum; the merging-taper L = W x S formula does not apply

One-lane two-way taper = 100 ft (Sec 6B.08 50-100 ft band)

Required: 100 ft (one-lane two-way taper, 50-100 ft band per MUTCD Sec 6B.08; plan uses the 100 ft maximum)

Field	Value
Closure type	lane
Offset (lane width)	10.5 ft
Required taper (one-lane two-way taper)	100 ft

Source: MUTCD 11th Ed. Sec 6B.08 (one-lane, two-way traffic control): 50-100 ft taper with channelizing devices at approximately 20 ft spacing. CDOT S-630-1 Case 17 warns the taper must be short enough to not be mistaken for a transition.

CDOT reference: MUTCD 11th Ed. Part 6 TA-10 (flagger-controlled one-lane two-way operation)

Buffer Space

MUTCD Table 6B-2: 305 ft (CDOT supplement silent at this speed)

Source: MUTCD 11th Ed. Sec 6B.06, Table 6B-2 (stopping sight distance)

Channelizing Device Spacing

20 ft spacing in the one-lane two-way taper (MUTCD Sec 6B.08: approximately 20 ft, taper-specific guidance overriding the Sec 6K.01 speed-based rule)

40 mph -> 80 ft spacing (MUTCD Sec 6K.01: spacing equals 2x speed in feet)

one-lane two-way taper = 100 ft / 20 ft max spacing -> 6 drums at 20.0 ft intervals

3000 ft / 80 ft max spacing -> 39 cones at 78.9 ft intervals

Downstream taper: 50 ft at ~12.5 ft spacing -> 4 cones (MUTCD Sec 6B.08: 50-100 ft downstream taper); 39 tangent + 4 downstream = 43 cones total

Source: MUTCD 11th Ed. Sec 6B.08 (one-lane two-way taper) + Sec 6K.01 (tangent)

Advance Warning Signs

Speed 40 mph, road_type = 'urban_low' -> Table 6B-1 category: urban_low

A = 100 ft, B = 100 ft, C = 100 ft

Position	Code	Station (ft)	Distance from Taper (ft)
Advance (S1-1)	S1-1	4,405	1,000 upstream
C (furthest)	W20-1	3,905	500 upstream
B (middle)	W20-4	3,805	400 upstream
between A and B (TA-10 note 8)	W3-4	3,705	300 upstream
Advance (W20-7a)	W20-7a	3,605	200 upstream

Source: MUTCD 11th Ed. Table 6B-1

Colorado Requirements (CDOT S-630-1)

Check	Result	Citation	Detail
Signs on both sides of divided highway	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 8	Required: False. Signs placed: 7 left, 13 right.
G20-5P Work Zone signs every 2,640 ft	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 4	Zone length: 3,905 ft. Required: 2. Placed: 2.
Speed reduction <= 15 mph per sign installation	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 3	No work-zone speed reduction. Posted speed 40 mph applies throughout the zone.
Flagger station lighting 500W @ 8 ft	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 22	Not applicable (no flaggers).

AADT threshold for mobile operations (<= 2,000) — Not applicable (not a mobile operation). (CDOT S-630-1 (July 2026) Sheet 23, Case 38 Note 1)

All Colorado checks pass: True

CDOT Case Reference

MUTCD TA-10: Flagger one-lane two-way

This scenario matches MUTCD 11th Ed. Part 6 TA-10 (the federal standard for flagger-controlled alternating one-way traffic on a 2-lane undivided highway). CDOT S-630-1 does not include a general flagger one-lane two-way case; Case 17 (lane closure at a curve) is the closest CDOT analog but is curve-specialized.

The generated plan follows the same device layout as the reference case with spacing computed for 40 mph.

Reference:
https://www.codot.gov/safety/traffic-safety/assets/s-standard-plans/s-630-1/S-630-01%20(19-Page%20Set).pdf

Geometry Validation

Work zone 3000 ft vs required taper 100 ft (one-lane two-way taper) + buffer 305 ft.

All geometry checks pass: True

Source: MUTCD 11th Ed. Sec 6B.06 (buffer) and Sec 6B.08 (taper)

Corridor Validation

Corridor check not run (no site coordinates supplied).

Site Adjustments

Site-condition flags layered onto the baseline MUTCD/CDOT layout. Each adjustment is traced to its source rule; the rendered plan, device list, and crew narrative reflect every item below.

Rule	Adjustment
MUTCD §6N.12 p. 848 — Work within the Traveled Way at an Intersection (11th Ed.)	No devices added — the cross-street approach layout is not generated; see the pending-verification disclosure.
MUTCD §6N.16 p. 851 — Interchanges (11th Ed.)	No devices added — the per-ramp interchange layout is not generated; see the pending-verification disclosure.
MUTCD §6C.02 — pedestrian considerations in work zones	Added 4 Type III barricades (sidewalk closure points) and 2 R9-9 SIDEWALK CLOSED signs at the upstream and downstream ends of the work zone.
MUTCD §7B.08 — school zone signing in proximity to work zones	Added 2 S1-1 SCHOOL signs upstream of the advance warning signs (station 4,405 ft, both sides).

Flagger Control

Two flagger stations are required for alternating one-way operations: one 100 ft upstream of the one-lane two-way taper (MUTCD Fig. 6P-10 50-100 ft band) to stop right-direction traffic, one 300 ft past the work-area end (CDOT S-630-1 Case 17 "200' TO 300'" standoff) to stop opposing traffic. Flagger stations should be visible to approaching drivers from the full advance-warning distance.

Source: MUTCD 11th Ed. Chapter 6D (Flagger Control) and Chapter 6E (One-Lane, Two-Way Traffic Control).

Pending Verification

2 reference(s) pending verification. An adjacent intersection is flagged, but Conestruct does not generate the cross-street approach layout or evaluate traffic-signal operation, and adds no devices for it. Cross-street control design and the signal-operation review (MUTCD §6N.12, 11th Ed. p. 848; Part 4 for signal-head visibility) remain with the traffic control supervisor. Intersection support is tracked at issue #117; corner-quadrant work at issue #128. Tracking: https://github.com/rtmakatura/conestruct/issues/117

- [intersection_layout_not_generated] An adjacent intersection is flagged, but Conestruct does not generate the cross-street approach layout or evaluate traffic-signal operation, and adds no devices for it. Cross-street control design and the signal-operation review (MUTCD §6N.12, 11th Ed. p. 848; Part 4 for signal-head visibility) remain with the traffic control supervisor. Intersection support is tracked at issue #117; corner-quadrant work at issue #128. (https://github.com/rtmakatura/conestruct/issues/117)
- [interchange_layout_not_generated] An adjacent interchange is flagged, but Conestruct does not generate the per-ramp layout and adds no devices for it. Ramp-specific control design and the ramp-access and coordination review (MUTCD §6N.16, 11th Ed. p. 851) remain with the traffic control supervisor. Tracked at issue #117. (https://github.com/rtmakatura/conestruct/issues/117)